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# How to Survive a Mass Extinction: Lessons from Fern Gametophytes

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Abstract.—Ferns are one of the most diverse lineages of land plants and are crucial members of many of Earth's ecosystems, playing roles such as ecological indicators or key players in nutrient cycling. These roles, however, have largely been studied in the sporophyte generation; the gametophyte generation, which is the sexual phase of the life cycle and therefore the establishment point for new populations, is relatively understudied. Understanding the role of the gametophyte during periods of environmental change is therefore critical for understanding fern responses more broadly. Perhaps the most extreme example of fern persistence during and after extreme environmental change lies in the fossil record immediately following the Chicxulub asteroid

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impact 66 million years ago. Ferns were among the first terrestrial flora to re-establish after the Cretaceous-Paleogene (K-Pg) mass extinction event, as evidenced by the "fern spike" or "fern spore spike" found globally in the palynological record from this time. To evaluate the effects of post-impact conditions on fern gametophytes, we compared spore germination rates and physiological performance in control conditions with experimental conditions analogous to those following the Chicxulub impact (*i.e.*, elevated $CO_2$, decreased temperature, decreased light, acidic precipitation). We found that spores in post-impact conditions took longer to germinate but gametophytes had significantly higher photosynthetic yield, indicating that the simulated conditions did not hinder their success. Our experiment used a multi-factorial approach to determine the ability of fern spores and gametophytes to germinate and function in a combination of stressful conditions, and our results suggest that fern gametophytes are entirely capable of persisting in the environment of the Early Paleogene. Further studies will be necessary to tease apart the effects of individual environmental factors, but we present a starting point for understanding ferns' responses to extreme conditions and the fern spore spike phenomenon.

Key Words.—Extinction, fern, gametophyte, physiology

Ferns are important members of many of Earth's ecosystems, and with ca. 10,000 species are one of the most diverse lineages of land plants (PPG I, 2016). Ferns have been shown to play various roles in ecosystems, including as ecological indicators (Della, 2022; Duque *et al.*, 2005; Zuquim *et al.*, 2014), facilitators of other species (Azevedo-Schmidt *et al.*, 2024; Gallegos *et al.*, 2015; Wan *et al.*, 2014), and key players in nutrient cycling (Cardelús and Mack, 2010) and other soil processes (Sanchez-Castillo *et al.*, 2019; Zhao *et al.*, 2012). The role of ferns as pioneer species in new habitats has also long been recognized (Thomas and Cleal, 2021): ferns are often among the first species to colonize new substrates following disturbance events (Barrington, 1993) and are disproportionately overrepresented in the floras of oceanic islands (Moran, 2008; Perrie and Brownsey, 2007; Tryon, 1970). In addition to their success at long-distance dispersal, which is facilitated by their minute, dust-like spores (which in some species can remain viable for decades [Berry, 2019; Nayar *et al.*, 1971; Spicer and Collinson, 2014]), ferns are also physiologically resilient, often far more so than they are assumed to be. Fern gametophytes can be long-lived, with documented longevity ranging from four to more than 25 months (Watkins, Mack, and Mulkey, 2007), and desiccation tolerant (Watkins, Holbrook, and Zwieniecki, 2010; Watkins *et al.*, 2007). Fern sporophytes have evolved an array of defensive, insecticidal proteins (Liu *et al.*, 2019; Wei *et al.*, 2023), and can be tolerant of heavy metals (Chang, Yoon, and Kim, 2009; Schmitt *et al.*, 2017), to the extent that some species are even used as phytoremediators for soil toxins like arsenic (Praveen and Pandey, 2020).

The resilience of ferns is perhaps best exemplified by their survival and rise to prominence following the Chicxulub asteroid impact and resulting Cretaceous-Paleogene (K-Pg) mass extinction event that occurred 66 million years ago (mya). The Chicxulub impact and its aftermath caused the extinction of approximately 75% of all plant and animal species on Earth (Jablonski, 1994). The harsh post-impact environment resembled nuclear winter and consisted of dim, cool, and acidic conditions (Bardeen *et al.*, 2017; Chiarenza *et al.*, 2020; Engvild, 2003; Schulte *et al.*, 2010; V Vajda, Raine, and Hollis, 2001; Vellekoop *et al.*, 2015). Global fires contributed to an abundance of soot and smoke, reducing the amount

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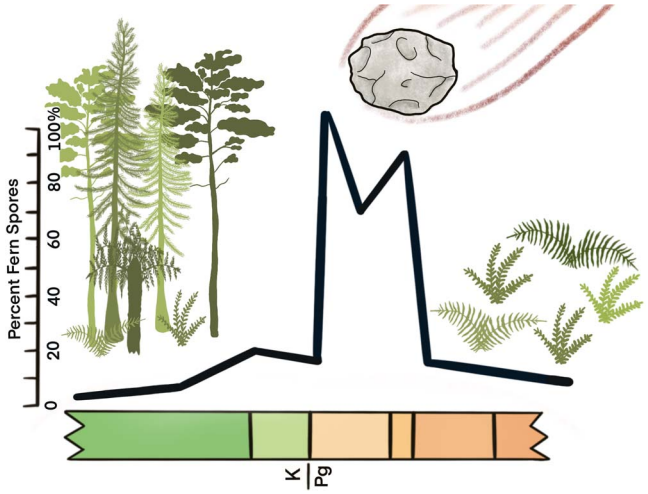



FIG. 1. Conceptual illustration of the percent of the palynological (pollen and spore) record comprised of fern spores across the Cretaceous-Paleogene (K-Pg) boundary, ca. 65 million years ago (data from Tschudy *et al.*, 1984). Prior to the asteroid impact at the boundary, rich and diverse floristic communities existed and ferns spores comprised up to 20% of the palynological record. Following the impact event, fern spores rose to above 70% and in some cases up to 100% of the palynological record, the phenomenon known as the "fern spore spike".

of light available and cooling the atmosphere. The impact released megatons of particulates, acidifying the air and contaminating the water with carbonates, sulfur, and other chemicals (Kring, 2007; Pope *et al.*, 1997; Schulte *et al.*, 2010). These hostile conditions presented a serious challenge for the planet's vegetation, and an estimated 57% of all plant species in western North America were driven to extinction (Labandeira, Johnson, and Wilf, 2002). The palynological record immediately post-impact is depauperate, with virtually no evidence of pollen or spores in the period immediately following the event (Orth *et al.*, 1981; Vivi Vajda and Bercovici, 2014). However, above the K-Pg impact debris layer, in the period representing the earliest recovery from the extinction event, fern spores occur in extraordinarily high relative abundance (Schultz and D'Hondt, 1996; Tschudy *et al.*, 1984). Commonly known as the "fern spike" or "fern spore spike", this feature in the fossil record is assumed to reflect a dramatic increase of ferns relative to other plants in the recovering terrestrial flora (Fig. 1). This phenomenon is so consistent that it is now recognized globally as a biomarker for the K-Pg boundary (Schulte *et al.*, 2010; Vajda and McLoughlin, 2007). However, how ferns were able to reestablish and thrive in harsh post-extinction environments remains unknown.

Explanations for the success of ferns above the K-Pg boundary highlight either their superior capacity to disperse and colonize, or various ecophysiological characteristics that would have fostered a competitive advantage in early Paleogene environments. The differences in dispersal ability between ferns and seed plants may have contributed to the success of ferns by allowing them to more

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rapidly re-colonize substrates or areas made available post-impact. Spores are generally much smaller and lighter than seeds; on average they measure 30-50 μm and have a mass of ca. 34 ng (Gómez-Noguez *et al.*, 2016). This allows spores to disperse longer distances, on average, as dispersal distance correlates with size (Howe and Smallwood, 1982), and they are dispersed primarily by wind (Moran, 2008). In an environment where animal dispersal agents were all but wiped out, ferns' dispersal characteristics may have given them an advantage over animal-dependent seed plants, and enabled ferns to spread more rapidly in habitats newly devoid of competition (Jones *et al.*, 2006; Tschudy *et al.*, 1984). However, given that ferns dominated the landscape for up to tens of thousands of years following the impact (Clyde *et al.*, 2016; Vajda, Raine, and Hollis, 2001), dispersal capacity alone appears insufficient to explain the fern spike; other mechanisms likely played a role in fern establishment and success in the Early Paleogene.

Ferns possess a unique suite of physiological traits that have allowed them collectively to diversify into numerous niche spaces and habitat types. Ferns can be found from deserts to rainforests and growing across substrates, ranging from fully submerged aquatics to epiphytes to serpentine specialists (Mehltreter, Walker, and Sharpe, 2010). Some of the characteristics that underlie this broad habitat versatility are also thought to have facilitated ferns' success as "disaster taxa" (Thomas and Cleal, 2021), including in post-impact conditions of the Early Paleogene. These include their ability to tolerate soils that are nutrient-poor, saline, anoxic, and/or contaminated with heavy metals (Husby *et al.*, 2011; Kachenko, Singh, and Bhatia, 2007; Tschudy *et al.*, 1984); their acquisition of a specialized photoreceptor (neochrome) that allows them to thrive in low light conditions (Kawai *et al.*, 2003; Li *et al.*, 2014); their capacity for drought resistance (Baer, Wheeler, and Pittermann, 2016; Holmlund *et al.*, 2019); and their establishment of root and rhizome systems that can grow rapidly and may be buried meters deep, which allows them to reach reservoirs of water and nutrients that other plants cannot (Hartig and Beck, 2003; Husby, 2013). Ferns also establish perennial spore banks that can contribute to rapid regrowth on competitor-free substrates (Berry, 2019; Dyer and Lindsay, 1992; Esaete *et al.*, 2014). Even polyploidy, which is extremely common in ferns (Wood *et al.*, 2009), has been suggested as playing a causal role in ferns' success post-impact, by enlarging their capacity for genetic and physiological plasticity via neofunctionalization (Berry, 2019; Fawcett, Maere, and Van de Peer, 2009).

With the exception of polyploidy, all of the traits cited above involve the sporophyte generation of the life cycle. In contrast, little attention has been paid to the gametophyte generation when considering the ecological success of ferns following environmental upheaval. Like all other land plants, ferns alternate between haploid gametophyte and diploid sporophyte life phases, but they differ from most other plants in that these phases are nutritionally independent from one another (Haufler *et al.*, 2016). The gametophyte germinates from a spore and upon reaching sexual maturity, a variety of sexual reproductive strategies are typically available, ranging from extreme inbreeding to outcrossing (Haufler *et al.*, 2016; Sessa, Testo, and Watkins, 2016). Regardless

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of the specific pathway followed, the production and establishment of a new sporophyte via sexual reproduction depends on successful fertilization at the gametophyte stage. While the gametophyte has historically been thought of as a vulnerability in the fern life cycle (Klekowski and Baker, 1966; Page, 2002), other studies have emphasized that the gametophyte can be more physiologically hardy than the sporophyte (Farrar *et al.*, 2008; Sato and Sakai, 1981; Watkins, Mack, and Mulkey, 2007; Watkins *et al.*, 2007), and can even occupy a larger physiological and ecological niche (Krieg et al., 2026; Krieg and Chambers, 2022; Nitta *et al.*, 2021; Pinson *et al.*, 2017; Sato and Sakai, 1981). Because of their critical role as the establishment point in the fern life cycle, determining the effects of post-impact conditions on gametophytes is essential for understanding the success of ferns after such events.

To explore the behavior of gametophytes and their potential role in ferns' success following the K-Pg mass extinction, we focused on the physiological impact of hypothesized Early Paleogene conditions on gametophytes. We selected species from two fern families (Blechnaceae and Cyatheaceae) that were extant at the time of and immediately following the Chicxulub impact, based on the fossil record (Collinson, 2001; Collinson, 1996) and molecular dating analyses (Pelosi *et al.*, 2022; Schuettpelz and Pryer, 2009; Testo and Sundue, 2016). We grew gametophytes in control and experimental environmental conditions analogous to the post mass extinction environment (elevated $CO_2$, decreased light, decreased temperature, and watering with a mildly acidic solution). We compared germination rates of spores and physiological performance (as measured by photosynthetic yield) of gametophytes in the experimental and control conditions to evaluate the extent to which the post-impact conditions would have affected ferns' survival in the gametophyte stage.

## Materials and Methods

Spores representing two lineages present at the time of the K-Pg boundary (Spicer and Collinson, 2014; Vajda and Bercovici, 2014) were used as our study species: *Blechnum occidentale* L. (Blechnaceae), *Sphaeropteris horrida* (Liebm.) R.M. Tryon (Cyatheaceae), and *Telmatoblechnum serrulatum* (Rich.) Perrie, D.J. Ohlsen & Brownsey (Blechnaceae) (Fig. 2).

To measure average number of days to germination, spores were sown on Petri dishes containing Bold's medium (Bold, 1957) with Nitsch's micronutrients (Nitsch, 1951) and placed in an I-36LL grown chamber (Percival Scientific, Perry, Iowa, USA) in control conditions: ambient $CO_2$ (ca. 500 ppm); moderate light (80 $\mu mol/m^2/s$ with a 12:12 light/dark cycle); ambient temperature (25°C); and weekly watering with deionized water. We observed plates weekly for eight weeks and noted whether spores had germinated after that period.

To assess physiological performance, spores were plated as above for a total of 14 plates per species in each of the control and experimental treatments (3 species $\times$ 14 plates $\times$ 2 treatments). The control conditions were as described above. In a separate I-36LL chamber, the experimental treatment consisted

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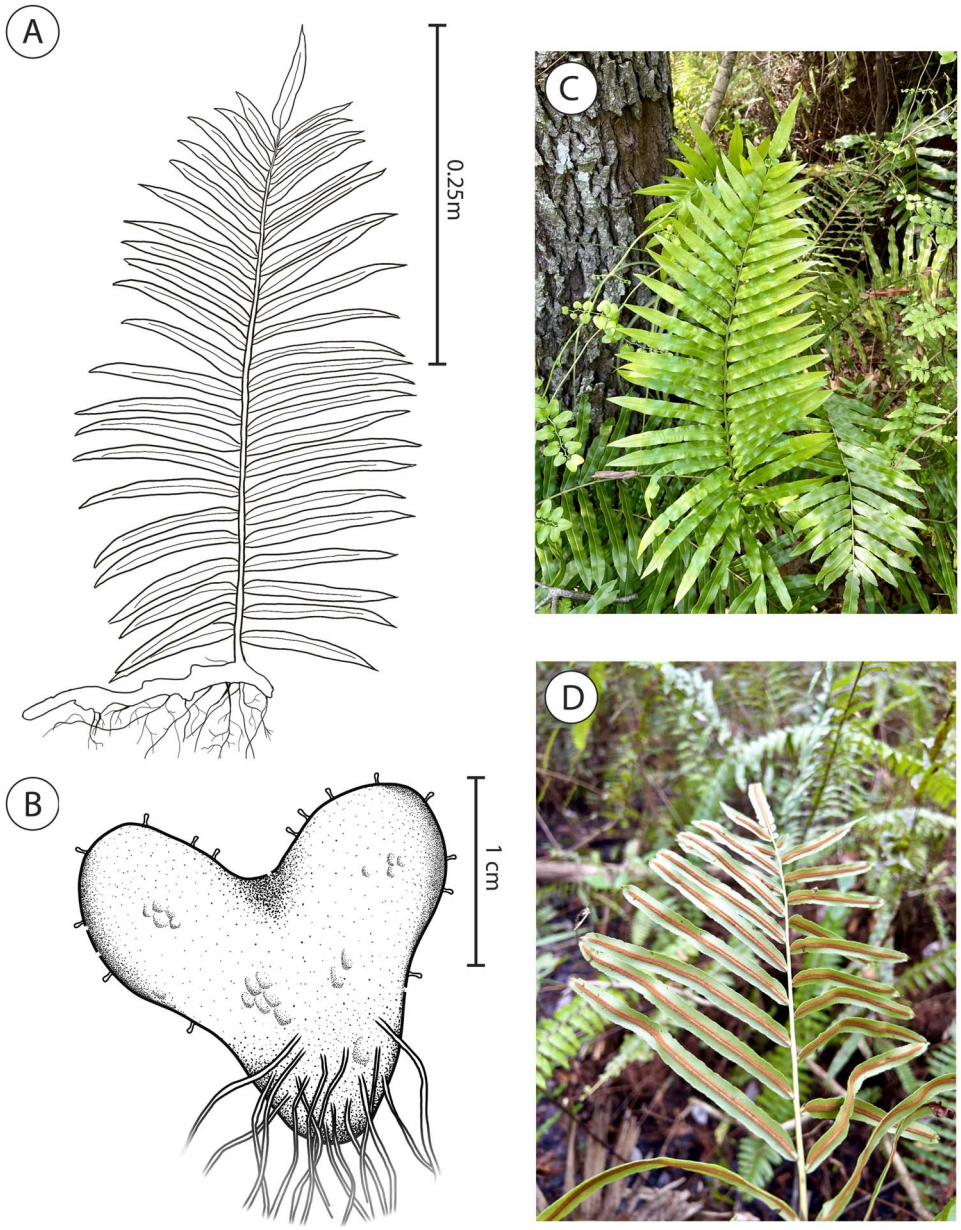



FIG. 2. Our study species, *Telmatoblechnum serrulatum*. Illustrations (A, B) and photographs (C, D) of sporophytes (A, C, D) and gametophyte (B). Photographs by E.B. Sessa; sporophyte illustration by K. Wheatley; gametophyte illustration by E. H. Kim.

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of: elevated $CO_2$ (1000 ppm); decreased light (16.92 ± 2.76 μmol/$m^2$/s); decreased temperature (19.50 ± 0.88°C); and weekly watering with a dilute sulfuric acid solution at pH 4.5. These conditions were chosen based on a survey of the literature estimating climatic dynamics in the early Paleogene (Bardeen *et al.*, 2017; Schulte *et al.*, 2010; Vellekoop *et al.*, 2015).

Of the 14 plates per species in each treatment, seven were placed with the lid up and seven with the lid down (to determine whether liquid pooling on the surface of the medium would impact spore germination or gametophyte growth). We found no difference in germination success or growth patterns between plates with differing orientations; therefore, data from both orientations were pooled for analysis. Plate locations were randomized within each growth chamber two to three times per week, while orientation of the lid remained constant. Gametophytes were grown to maturity, which was indicated by the presence of gametangia, and time to first germination on each plate was documented.

Measurements of the maximum quantum yield of photosystem II ($F_v/F_m$) were taken with a MiniPam II (Heinz Walz, Effeltrich, Germany) 65 days after sowing spores on sexually mature gametophytes. This value represents the ratio of the variable to maximum fluorescence and gives a quantitative evaluation of photosynthetic efficiency. This value is unitless and scales from 0–1, with measurements closer to 1 typically interpreted as a proxy for low stress, while values closer to 0 indicate heightened stress. This measurement is applicable to all photosynthesizing organisms and is calculated as follows (Maxwell and Johnson, 2000):

$$F_v/F_m = (F_m - F_0)/F_m$$

where $F_m$ is maximum fluorescence yield and $F_0$ is the minimal level of fluorescence.

To obtain $F_v/F_m$ measurements, plates were first placed in a sealed box without access to light for 15 minutes for dark acclimation. One gametophyte from each plate was then placed under the fluorometer at a distance of 3 mm and an angle of 45° and $F_v/F_m$ measurements were taken. Irregularities such as algal growth or gametophyte folding were documented. The plates were then returned to the dark box to allow neighboring gametophytes to reacclimate to the dark. After reacclimation, the procedure was repeated until all gametophytes were measured.

We used Wilcoxon rank sum tests with continuity correction to determine whether the control and treatment were significantly different in average number of days to germination. The same test was used for $F_v/F_m$ measurements. All statistical analyses were performed in R version 4.1.1 (R Core Team, 2021; all data and code available at https://github.com/ehk-kim/fernGametoEnv).

## Results

Of the three species evaluated, only *Telmatoblechnum serrulatum* germinated and reached maturity in both the control and treatment conditions in sufficient numbers to make comparable measurements and draw robust conclusions, and so results are presented only for this species. The average number of

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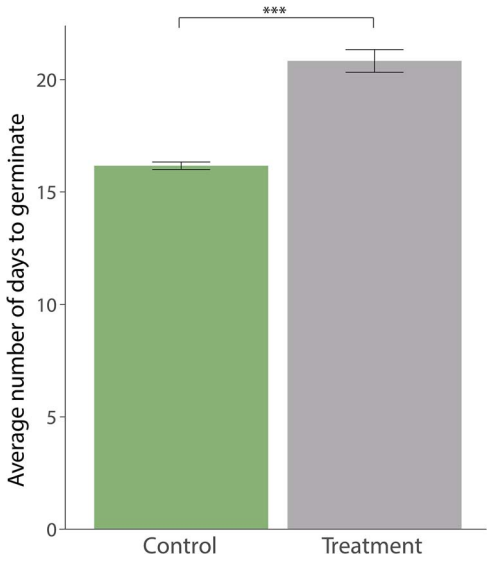



FIG. 3. Average number of days from plating to spore germination in the control (green) and treatment (gray) environments for *Telmatoblechnum serrulatum*. *** indicates $P < 0.001$.

days to germination was significantly different between the treatment and the control (Wilcoxon rank sum test with continuity correction, $P = 1.016 \times 10^{-5}$, $df = 74$), with the gametophytes in the experimental treatment taking longer to germinate than spores in the control conditions (control mean = 16.2 days, treatment mean = 20.8 days, Fig 3). Gametophytes in the experimental treatment had a significantly higher average $F_v/F_m$ value than those in the control, indicative of lower stress levels (control mean = 0.239, treatment mean = 0.474, Wilcoxon rank sum test with continuity correction, $P = 2.492 \times 10^{-8}$, $df = 74$; Fig. 4) (The authors note that given the relatively low $F_v/F_m$ values recovered for the control treatment, additional control experiments were conducted to confirm whether these values were repeatable or represented abnormally stressful conditions; the additional results did not differ from the original results, and gametophytes continued to grow in the growth chamber subsequently).

While only *Telmatoblechnum serrulatum* grew in both treatments, *Sphaeropteris horrida* spores did germinate in the Early Paleogene conditions but not in the control chamber. *Blechnum occidentale* spores germinated in both the control and treatment conditions but not in sufficient numbers for us to collect and analyze data.

## DISCUSSION

Ferns are a resilient clade of plants that survived and even thrived in conditions uninhabitable by many species following the Chicxulub impact event at the end of the Cretaceous period. Little is known, however, about the mechanisms by which ferns were able to persist in these conditions. The gametophyte is of particular

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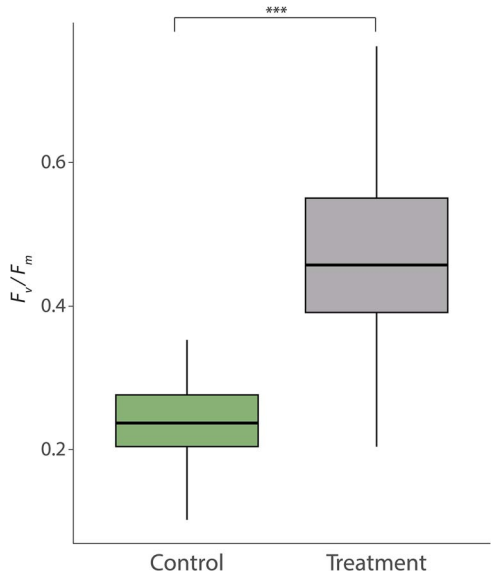



FIG. 4. Values of $F_v/F_m$ for mature *Telmatoblechnum serrulatum* gametophytes in the control (green) and treatment (gray) environments. *** indicates $P < 0.001$.

interest as the establishment of new fern populations is dependent on the success of the gametophyte generation, and the survival and development of fern populations therefore requires the gametophyte to respond effectively to environmental stressors. Given their importance in the life cycle, we evaluated the physiological effects of a simulated post-impact environment on the success of fern gametophytes to better understand ferns' ability to establish and persist in harsh environmental conditions where most plants could not.

Overall, we found that our post-impact environmental conditions did not appear to hinder the success of the gametophytes' photosynthetic efficiency. The average $F_v/F_m$ value for gametophytes in the mass extinction conditions was significantly higher than that of the control group, reflecting a relatively higher yield of photosystem II and less stress than in the control conditions. Interestingly, spores in the mass extinction conditions germinated later than those in the control group, suggesting that our experimental conditions did have some inhibitory effect on spore germination; once germination had occurred, however, and gametophytes reached maturity, they were clearly capable of coping with these conditions and even outperforming the individuals in the control conditions. The differences in gametophyte response and performance that we observed in this experiment cannot be explained by any one element of the treatment conditions, as we used a multi-factor experimental treatment intentionally in order to assess the total effect of the putative post-impact environment on gametophyte performance, but it is clear from our results that gametophytes are able to tolerate and function successfully in this combination of conditions. Below we discuss the expected responses of fern gametophytes to the individual

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stressors we employed (light, carbon dioxide, acid, and temperature) and evaluate our results in light of these previous findings.

Light intensity and quality are key factors that influence spore germination and development of gametophytes (Miller, 1968; Raghavan, 1980). According to Wada (2019) nearly all early developmental processes in the fern gametophyte are controlled by light. Similar to their sporophyte counterparts, gametophytes can employ mechanisms to adapt to varying light levels such as increasing concentrations of photoprotective pigments and altering the molecular pathways used in photosynthesis (Fernández-Marín *et al.*, 2012; Krieg and Chambers, 2022). In fact, lower light intensities may enhance spore germination (Isikawa and Oohusa, 1956; Marcondes-Ferreira and Felippe, 1984), and Nitta *et al.* (2021) suggested that gametophytes may be adapted to lower light levels compared to sporophytes. Perhaps one of the most important events in fern evolution was the acquisition of a bryophyte-derived neochrome photoreceptor via horizontal gene transfer (Li *et al.*, 2014). These pigments are suggested to have facilitated the diversification of ferns in low-light conditions such as angiosperm understories (Cai *et al.*, 2021). Furthermore, the expansion of CRY gene families in Cyatheales and Polypodiales has aided in the rapid response of stomata to blue-light (Cai *et al.*, 2021). The ability to promote stomatal activity in low-light conditions further explains the ability of ferns to occupy habitats beneath angiosperm-dominated canopies, and may have contributed to the rapid diversification of the Polypodiales during the Cretaceous, in "the shadow of angiosperms" (Schneider *et al.*, 2004). Dimly-lit forest understory conditions may be similar to those that followed the Chicxulub impact, and these adaptations to low-light conditions in angiosperm-dominated environments may also have facilitated the survival of ferns in post-impact environments.

Increased atmospheric $CO_2$ levels would almost certainly have followed the Chicxulub impact (Royer, Pagani, and Beerling, 2012). Coping with changing environmental concentrations of $CO_2$ is primarily accomplished through stomatal regulation in the sporophyte generation, and most studies of atmospheric $CO_2$ response in ferns and other plants have focused on that life stage. While our study was focused on gametophytes, it is worth noting that stomatal structure and function are not the same across vascular plants. The ferns (as well as the lycophytes and gymnosperms) primarily employ a simple, passive form of stomatal behavior, in contrast with angiosperms, which have more complex, environmentally-responsive stomatal control (Brodribb and McAdam, 2011). Unlike their sporophyte counterparts, however, fern gametophytes do not have stomata (Krieg and Chambers, 2022; Nitta *et al.*, 2021; Pinson *et al.*, 2017; Sato and Sakai, 1981) and must rely entirely on diffusion for gas exchange, a process aided by the typically thin nature of the gametophyte thallus, which is rarely more than a few cell layers thick. The very few studies of $CO_2$ variability on fern spores and gametophytes have demonstrated their ability to respond; for example, increases in $CO_2$ concentration have been shown to accelerate spore germination in *Pyrrosia piloselloides* (L.) Price (Ong, Koh, and Wee, 1998), but not in *Onoclea sensibilis* L. (Edwards, 1977). Gametophyte growth, sexual development, and production of sporophytes were also all accelerated in *P. piloselloides*

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with increased $CO_2$ concentration (Ong, Koh, and Wee, 1998). This trend is expected as increasing the available carbon should facilitate increased assimilation during photosynthesis. Interestingly, there is also the possibility for plasticity in the photosynthetic pathway utilized by each generation. For example, (Martin, Allen, and Haufler, 1995) found that C3 photosynthesis occurred in the gametophyte and immature sporophytes of *Pyrrosia longifolia* (Burm.f.) C. V. Morton, while mature sporophytes used CAM. How fern gametophytes respond to changes in $CO_2$ in general is worthy of much additional study, and whether differences in photosynthetic strategies may allow gametophytes to explore distinct niches compared to the sporophyte generation is a particularly interesting avenue for future research.

We also tested the effects of acidity on fern gametophytes, as acid rain was a likely component of the Early Paleogene environment (Schulte *et al.*, 2010; Sweet, 2001). While there have been limited studies on the effects of acid or simulated acid rain (SAR) on ferns, the available data are intriguing. When comparing the effects of SAR on several woody and herbaceous angiosperms and sporophytes of *Pteridium aquilinum* (L.) Kuhn, Evans and Curry (1979) found that *P. aquilinum* was the most sensitive of the plants tested. While all species incurred foliar damage from SAR application, primarily to their vascular tissues, it was most pronounced in the fern. In contrast, fern spores and gametophytes have been found to survive and even to reach optimal growth in acidic conditions (reviewed by Raghavan, 1989). For example, spores of the tree fern *Cyathea atrovirens* (Langsd. & Fisch.) Domin had higher germination rates in more acidic conditions (pH 4.0-6.5) compared to neutral pHs (Rechenmacher, Schmitt, and Droste, 2010). In some extreme cases, spore germination has been shown to be greatest at very low pH, as was observed to be the case for *Ophioglossum palmatum* L., which was able to germinate in media with a pH as low as 3.4 and had optimal germination at pH 4.0 (Whittier and Moyroud, 1993) (although highly acidic pH was found to slow growth of the rhizoid and protonemal initials [Courbet, 1955]). Using other species in a set of simulated acid rain experiments, Bosley, Petersen, and Rebbeck (1998) found that while spores of *Athyrium filix-femina* (L.) Roth and *Onoclea sensibilis* L. were able to germinate at low pH conditions (pH 4), there was a lower limit (pH 3) at which germination did not occur. In contrast, other ferns have been found to prefer more alkaline conditions, such as *Notholaena cochisensis* Goodd. and *Pellea limitanea* (Maxon) C. V. Morton, which occur naturally on limestone substrates with pH as high as 8-10 (Hevly, 1963). Compared to angiosperms, which typically germinate in optimal soil pHs around 6 (Haskell, 1951), the ability of fern spores to germinate and gametophytes to thrive in lower pHs may be an important aspect of their success in a variety of habitats and conditions, including in the harsh environment following the Chicxulub impact. Further experimental work on responses to simulated acid rain across the fern phylogeny will be required to fully assess the thresholds of spore and gametophyte tolerance, and would be highly informative both for understanding how ferns survived in the early Paleogene and how they may be expected to respond to increasing levels of acid rain in the future.

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Spore germination is, in part, dependent on temperature, and while the majority of individual fern species inhabit tropical regions, the group as a whole is globally distributed (Moran, 2008). In general, spore germination rates have been found to be optimal between 20°C-25°C (Raghavan, 1989), though optimal temperatures will vary depending on the species and its natural history. Germination in some species can occur at temperatures as low as 1-5°C (Conway, 1949) and spores can remain viable after prolonged exposure to cold temperatures (Ballesteros, Hill, and Walters, 2017; Hutchinson and Langeland, 2014). Gametophytes of temperate species can withstand severe freezing, far past the tolerance of the sporophyte (Farrar, 1978; Sato and Sakai, 1981). For example, *Crepidomanes intricatum* (Farrar) Ebihara & Weakley, a gametophyte-only species (reviewed in Pinson *et al.* [2017]) that grows in temperate Eastern North America and frequently experiences below-freezing temperatures in the winter, showed no damage after being subjected to temperatures as low as −11°C (Farrar, 1998). In contrast, closely related sporophytes showed severe but recoverable damage. Work with the vining fern *Lygodium microphyllum* (Cav.) R.Br. further highlights the potential for differential responses to freezing between the life stages of ferns: the gametophytes of this tropical species are able to recover from short-term freezing at −20°C, while fronds of the sporophyte are not (Pelosi *et al.*, 2025) (however, other authors have found sporophytes to be freeze-tolerant in other, typically temperate, taxa [see, *e.g.*, Fernández-Marín *et al.,* 2021]). Based on the findings of these previous studies, it seems likely that the decreased temperatures we used in our experimental treatments would not have been a barrier to successful germination and physiological functioning of fern gametophytes, in line with what we observed.

This study sought to evaluate the responses of fern gametophytes under climatic conditions analogous to the Early Paleogene environment on Earth, immediately following one of the largest environmental and mass extinction crises our planet has experienced. The success of ferns in the face of that upheaval is remarkable and worthy of study, in order to understand both how the planet's flora responded during a period of past environmental crisis and how we may expect it to do so as the current climate crisis unfolds. We shed light on the overall physiological responses of fern gametophytes to a suite of environmental stressors, and while we cannot yet definitively explain the fern spike with these results, our findings show that the gametophyte phase was likely not a limiting factor for ferns in the impact winter environments of the earliest Paleogene. Further studies will be necessary to understand better the specific mechanisms governing gametophyte response to these environmental factors (light, carbon dioxide, acid, and temperature), singly and in various combinations, and across a wider range of taxa. However, we have shown that in at least one species, this generation of the life cycle would have been capable of germination and physiological success in the conditions we assume to have been present in the Early Paleogene. Understanding stress response processes may enable us to use ferns in the recovery of damaged habitats (*e.g.*, Cardelús and Mack, 2010; Praveen and Pandey, 2020; Wan *et al.*, 2014), to predict more accurately how these plants will adapt to a changing climate (Karichu *et al.*, 2022; Kessler and Kluge, 2022; Testo and Watkins,

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2013), and potentially even how they will respond to space flight and extra-planetary colonization (Poulet, Fontaine, and Dussap, 2016; Salmi and Roux, 2008).

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